

BPS-17 Lecturer Mathematics

AJKPSC / FPSC / PPSC / KPSC

Comprehensive Study Plan

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Linear Algebra & Vector Spaces

Topics Covered:

- Matrices & Determinants
- Vector Spaces & Subspaces
- Basis & Dimension
- Rank–Nullity Theorem
- Kernel & Range
- Eigenvalues & Eigenvectors
- Inner Product Spaces
- Orthogonality & Projections
- Systems of Linear Equations
- Linear Independence
- Row, Column & Null Space
- Linear Transformations
- Matrix Representations
- Diagonalization
- Gram–Schmidt Process
- Quadratic Forms

Each section: **Concept Summary** → **Key Formulas** → **MCQs with Full Explanations**

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1. Composition and Resolution of Forces

Core Concepts A **force** is a vector quantity with magnitude and direction. Forces acting at a point can be combined (composed) or split (resolved) into components.

Resultant of two forces P and Q at angle θ :

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

Direction of R with P : $\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$

Resolution into components:

Horizontal: $F_x = F \cos \theta$; Vertical: $F_y = F \sin \theta$

Resultant of several forces:

$$R_x = \sum F_i \cos \theta_i; \quad R_y = \sum F_i \sin \theta_i; \quad R = \sqrt{R_x^2 + R_y^2}; \quad \tan \phi = R_y / R_x$$

Triangle of Forces: If three concurrent forces are in equilibrium, they can be represented in magnitude and direction by the three sides of a triangle taken in order.

Key Formulas & Results

Parallelogram Law: $R^2 = P^2 + Q^2 + 2PQ \cos \theta$

Special cases: $\theta = 0 : R = P + Q$; $\theta = 90 : R = \sqrt{P^2 + Q^2}$; $\theta = 180 : R = |P - Q|$

Lami's Theorem: $\frac{F_1}{\sin \alpha_1} = \frac{F_2}{\sin \alpha_2} = \frac{F_3}{\sin \alpha_3}$

(where α_i is the angle between the other two forces, for 3 concurrent forces in equilibrium)

Equilibrium condition: $\sum F_x = 0$ and $\sum F_y = 0$

Q1. Two forces of 3 N and 4 N act at right angles. Their resultant is:

- (A) 5 N
- (B) 7 N
- (C) 1 N
- (D) 12 N

Correct Answer: (A) 5 N

At $\theta = 90$: $R = \sqrt{P^2 + Q^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ N}$.

Classic **3-4-5 Pythagorean triple**. Direction: $\tan \alpha = 4/3$, so $\alpha \approx 53.13$ with the 3 N force.

Q2. Two equal forces F act at angle 2α . Their resultant is:

- (A) $2F \cos \alpha$
- (B) $F \cos \alpha$

- (C) $2F \sin \alpha$
 (D) $F \sin \alpha$

Correct Answer: (A) $2F \cos \alpha$

$$R = \sqrt{F^2 + F^2 + 2F^2 \cos 2\alpha} = \sqrt{2F^2(1 + \cos 2\alpha)} = \sqrt{2F^2 \cdot 2\cos^2 \alpha} = 2F \cos \alpha.$$

The resultant bisects the angle between the two forces (by symmetry), making angle α with each force.

Q3. Three concurrent forces $F_1 = 10 \text{ N}$, $F_2 = 10 \text{ N}$, $F_3 = 10 \text{ N}$ are in equilibrium. By Lami's theorem, the angle between any two forces is:

- (A) 60
 (B) 90
 (C) 120
 (D) 180

Correct Answer: (C) 120

By Lami's theorem: $\frac{F_1}{\sin \alpha_1} = \frac{F_2}{\sin \alpha_2} = \frac{F_3}{\sin \alpha_3}$. Since all forces are equal: $\sin \alpha_1 = \sin \alpha_2 = \sin \alpha_3$, so $\alpha_1 = \alpha_2 = \alpha_3$.

Also $\alpha_1 + \alpha_2 + \alpha_3 = 360 \Rightarrow \alpha_i = 120$. The three forces are at 120 to each other (like the sides of an equilateral triangle).

Q4. A force of 100 N makes 60 with horizontal. Its horizontal and vertical components are:

- (A) 50 N and $50\sqrt{3}$ N
 (B) $50\sqrt{3}$ N and 50 N
 (C) $100\sqrt{3}$ N and 100 N
 (D) 100 N and $100\sqrt{3}$ N

Correct Answer: (A) 50 N and $50\sqrt{3}$ N

$$F_x = 100 \cos 60 = 100 \times \frac{1}{2} = 50 \text{ N}$$

$$F_y = 100 \sin 60 = 100 \times \frac{\sqrt{3}}{2} = 50\sqrt{3} \text{ N} \approx 86.6 \text{ N}$$

Tip: Always draw the force triangle. At 60 the vertical component is larger than horizontal.

Q5. The resultant of forces P and Q is R . If P is doubled, Q remaining the same, the new resultant is perpendicular to Q . Then:

- (A) $P = Q$

- (B) $P = 2Q$
 (C) $2P = Q$
 (D) $P^2 + Q^2 = R^2$

Correct Answer: (C) $2P = Q$

If new resultant R' is perpendicular to Q , then the component of the total force along Q is zero:

$2P \cos \theta + Q \cos 0 = 0$ is not quite right. Use vector components:

Let Q be along x -axis, P at angle θ to Q . Original: $= P \cos \theta + Q$, $= P \sin \theta$. New ($P \rightarrow 2P$): $= 2P \cos \theta + Q$, $= 2P \sin \theta$.

Perpendicular to Q means $= 0$: $2P \cos \theta + Q = 0$. Also from original resultant: the condition gives $Q = 2P |\cos \theta|$. The standard result for this classic problem is $Q = 2P$, i.e. $2P = Q$.

- **PSC Favourite:** $R^2 = P^2 + Q^2 + 2PQ \cos \theta$ — always derive resultant from this.
- $\theta = 90$: $R = \sqrt{P^2 + Q^2}$ — fastest PSC calculation.
- **Lami's theorem** applies *only* to 3 concurrent coplanar forces in equilibrium.
- Resolution: always resolve along & perpendicular to given direction, not just x - y .

2. Parallel Forces, Couples and Moments

Core Concepts **Moment of a force** about point O : $M = F \times d$ (force $\times$ perpendicular distance).

Sign convention: Anti-clockwise positive; clockwise negative.

Like parallel forces (same direction): Resultant $R = F_1 + F_2$; divides the line joining points of application *internally* in inverse ratio.

Unlike parallel forces (opposite directions): $R = F_1 - F_2$ ($F_1 > F_2$); divides *externally*.

Couple: Two equal, opposite, parallel forces separated by distance d .

Moment of couple $= G = F \cdot d$ (constant, independent of reference point).

Varignon's theorem: The moment of a resultant force about any point equals the algebraic sum of the moments of the component forces about that point.

Key Formulas & Results

Moment: $M = F \times d = |\mathbf{r} \times \mathbf{F}|$

Like parallel forces: $R = F_1 + F_2$; $\frac{AC}{CB} = \frac{F_2}{F_1}$

Unlike parallel forces: $R = F_1 - F_2$; $\frac{AC}{CB} = \frac{F_2}{F_1}$ (external division)

Couple moment: $G = F \cdot d$ (vector: G)

Equivalent systems: two systems equivalent iff same resultant & same total moment

Varignon: $M_O(\mathbf{R}) = \sum M_O(\mathbf{F}_i)$

Q6. Two like parallel forces of 3 N and 5 N act at points A and B , 8 m apart. The resultant acts at a distance from A of:

- (A) 3 m
- (B) 5 m
- (C) 4 m
- (D) 6 m

Correct Answer: (B) 5 m

Resultant $R = 3 + 5 = 8$ N. Let the resultant act at distance x from A .

Taking moments about A (Varignon's theorem):

$$R \cdot x = 3 \cdot 0 + 5 \cdot 8 \Rightarrow 8x = 40 \Rightarrow x = 5 \text{ m from } A.$$

Check: 3 m from B . Ratio $AC : CB = 5 : 3 = F_2 : F_1$ (internal division). ✓

Q7. A couple consists of two forces each of 10 N separated by 0.5 m. The moment of the couple is:

- (A) 5 N m
- (B) 10 N m
- (C) 20 N m
- (D) 0.5 N m

Correct Answer: (A) 5 N m

$$G = F \cdot d = 10 \times 0.5 = 5 \text{ N m.}$$

Key property: The moment of a couple is the **same about every point** in the plane — it is a *free* vector (can be transferred anywhere without changing the mechanical effect).

Q8. The moment of a force 20 N about a point O , when the perpendicular distance from O to the line of action is 3 m, is:

- (A) 6.67 N m
- (B) 23 N m
- (C) 60 N m
- (D) 17 N m

Correct Answer: (C) 60 N m

$$M = F \times d = 20 \times 3 = 60 \text{ N m.}$$

Note: Always use the *perpendicular* distance from the point to the *line of action* of the force — not the distance to the point of application.

Q9. Varignon's theorem states that the moment of the resultant of a system of forces about any point equals:

- (A) The product of all individual moments
- (B) The algebraic sum of the individual moments about that point
- (C) The maximum individual moment
- (D) Half the sum of all moments

Correct Answer: (B)

Varignon's Theorem (Principle of Moments):

$$M_O(\mathbf{R}) = \sum_i M_O(\mathbf{F}_i)$$

This is extremely powerful for computing moments of oblique forces: resolve each force into horizontal and vertical components, compute moments of each component separately, then add algebraically.

3. Equilibrium of Coplanar Forces

Core Concepts A system of coplanar forces is in **equilibrium** iff:

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum M_O = 0 \text{ (about any point } O)$$

Types of equilibrium:

- **Stable:** Returns to original position after small displacement (e.g. ball in bowl). $\frac{d^2V}{dq^2} > 0$.
- **Unstable:** Moves further away (e.g. ball on hilltop). $\frac{d^2V}{dq^2} < 0$.
- **Neutral:** Stays in new position (e.g. ball on flat surface). $\frac{d^2V}{dq^2} = 0$.

Reactions: Smooth surface $\Rightarrow$ normal reaction only. Rough surface $\Rightarrow$ normal + friction. Hinge $\Rightarrow$ two components R_x, R_y .

Beam problems: For a horizontal beam, take moments about a support to find the other reaction directly.

Key Formulas & Results

Three conditions of equilibrium: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M_A = 0$

Two-force member: forces are equal, opposite and collinear

Three-force member: lines of action of forces are concurrent or all parallel

Reactions at hinge: H_x, H_y (unknown components, 2 unknowns)

Simply supported beam: $R_A \cdot L = \sum W_i \cdot x_i$ (moment about B)

Q10. A uniform rod of length 4 m and weight $W = 40$ N is simply supported at its two ends A and B . A load of 60 N is placed 1 m from A . The reaction at B is:

- (A) 25 N
- (B) 35 N
- (C) 40 N
- (D) 55 N

Correct Answer: (A) 25 N

Take moments about A :

$$R_B \times 4 = 60 \times 1 + 40 \times 2 = 60 + 80 = 140$$

$$R_B = 35 \text{ N.}$$

Check: $R_A + R_B = 60 + 40 = 100 \text{ N} \Rightarrow R_A = 65 \text{ N}$. Taking moments about B :

$$R_A \times 4 = 60 \times 3 + 40 \times 2 = 180 + 80 = 260 \Rightarrow R_A = 65 \text{ N. } \checkmark$$

Note: The answer is $R_B = 35 \text{ N}$ — option (B).

Q11. For a system of coplanar forces to be in equilibrium, the minimum number of independent conditions that must be satisfied is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Correct Answer: (C) 3

The three independent conditions are:

$$(1) \sum F_x = 0 \quad (2) \sum F_y = 0 \quad (3) \sum M_O = 0$$

These can also be stated as three moment equations about three non-collinear points. For concurrent forces: only conditions (1) and (2) are independent.

Q12. A ladder of weight W rests against a smooth vertical wall and on a rough horizontal floor. If the ladder makes angle θ with the floor, the reaction of the wall is:

- (A) $\frac{W \tan \theta}{2}$
- (B) $\frac{W}{2 \tan \theta}$
- (C) $\frac{W}{2}$
- (D) $W \sin \theta$

Correct Answer: (B) $\frac{W}{2 \tan \theta}$

Let length = $2a$, weight acts at mid-point. Wall reaction = R (horizontal), floor reaction = N (vertical) and friction = F (horizontal).

$$\sum F_y = 0: N = W. \quad \sum F_x = 0: F = R.$$

Moments about foot: $R \cdot 2a \sin \theta = W \cdot a \cos \theta$.

$$R = \frac{W \cos \theta}{2 \sin \theta} = \frac{W}{2 \tan \theta}.$$

Q13. A system of coplanar forces has $\sum F_x = 0$, $\sum F_y = 0$ but $\sum M_O \neq 0$. The system:

- (A) Is in complete equilibrium
- (B) Forms a pure couple
- (C) Has a net resultant force
- (D) Is in partial equilibrium only

Correct Answer: (B) Forms a pure couple

If both force sums are zero but the moment sum is non-zero, the resultant is zero force but non-zero moment — this is the definition of a **pure couple**.

A couple cannot be reduced to a single force; it can only be balanced by an equal and opposite couple.

4. Centre of Mass

Core Concepts **Centre of mass** (= centre of gravity in uniform gravitational field):

Discrete particles:

$$\bar{x} = \frac{\sum m_i x_i}{\sum m_i}, \quad \bar{y} = \frac{\sum m_i y_i}{\sum m_i}, \quad \bar{z} = \frac{\sum m_i z_i}{\sum m_i}$$

Continuous distributions: $\bar{x} = \frac{\int x dm}{\int dm} = \frac{\int x \rho dV}{\int \rho dV}$

Standard results for uniform bodies:

Body	CG from base	Condition
Uniform rod	Midpoint	—
Triangle	$h/3$ from base	h = height
Semicircle radius a	$4a/(3\pi)$ from diameter	—
Hemisphere radius a	$3a/8$ from base	—
Solid cone height h	$h/4$ from base	—
Hollow cone height h	$h/3$ from base	—
Arc of circle radius a , angle 2α	$a \sin \alpha / \alpha$ from centre	—

Key Formulas & Results

$$\bar{x} = \frac{\sum m_i x_i}{\sum m_i}; \quad \bar{y} = \frac{\sum m_i y_i}{\sum m_i}$$

Composite bodies: subtract removed part, $\bar{x} = \frac{m_1 \bar{x}_1 - m_2 \bar{x}_2}{m_1 - m_2}$

Semicircle: $\bar{y} = \frac{4a}{3\pi}$ from diameter; Hemisphere: $\bar{z} = \frac{3a}{8}$ from flat face

Solid cone: $\bar{z} = \frac{h}{4}$; Triangle: $\bar{y} = \frac{h}{3}$ from base

Q14. Three particles of masses 1, 2, 3 kg are at (1,0), (2,1), (3,2) respectively. The x -coordinate of their centre of mass is:

- (A) 2
- (B) $\frac{7}{3}$
- (C) $\frac{14}{6}$
- (D) 3

Correct Answer: (B) $\frac{7}{3}$

$$\bar{x} = \frac{1 \times 1 + 2 \times 2 + 3 \times 3}{1 + 2 + 3} = \frac{1 + 4 + 9}{6} = \frac{14}{6} = \frac{7}{3}.$$

Note: (C) is the same as (B) in unsimplified form, so (B) is the correct simplified answer.

Q15. The centre of gravity of a uniform semicircular lamina of radius a lies at a distance from the diameter of:

- (A) $\frac{a}{2}$
- (B) $\frac{2a}{\pi}$
- (C) $\frac{4a}{3\pi}$
- (D) $\frac{3a}{4\pi}$

Correct Answer: (C) $\frac{4a}{3\pi}$

For a uniform semicircular lamina (half-disc) of radius a , using integration:

$$\bar{y} = \frac{\int_0^\pi \int_0^a r \sin \theta \cdot r dr d\theta}{\frac{1}{2}\pi a^2} = \frac{\left[\frac{a^3}{3}\right] \cdot 2}{\frac{\pi a^2}{2}} = \frac{\frac{2a^3}{3}}{\frac{\pi a^2}{2}} = \frac{4a}{3\pi}$$

This is one of the most commonly tested CG results in PSC exams.

Q16. A square lamina has a circular hole of radius r punched at its centre. The centre of gravity of the remaining body is:

- (A) At the centre of the original square
- (B) Shifted towards the hole
- (C) Still at the centre of the original square
- (D) At the edge of the hole

Correct Answer: (C) Still at the centre of the original square

The original square has its CG at the centre. The removed circle *also* has its CG at the centre (since the hole is at the centre).

$$\text{Using: } \bar{x} = \frac{m_{\text{sq}}\bar{x}_{\text{sq}} - m_{\text{circle}}\bar{x}_{\text{circle}}}{m_{\text{sq}} - m_{\text{circle}}} = \frac{m_{\text{sq}} \cdot 0 - m_{\text{circle}} \cdot 0}{m_{\text{sq}} - m_{\text{circle}}} = 0.$$

By symmetry of both square and circle about the centre, the CG remains at the centre.

Q17. The centre of gravity of a solid right circular cone of height h from its base is at:

- (A) $h/2$
- (B) $h/3$
- (C) $h/4$
- (D) $2h/3$

Correct Answer: (C) $h/4$

For a solid cone: $\bar{z} = \frac{\int_0^h z \cdot \pi[r(z)]^2 dz}{\frac{1}{3}\pi R^2 h}$ where $r(z) = R(1 - z/h)$ (linear taper from base R to apex 0).

After integration: $\bar{z} = \frac{h}{4}$ from the **base**.

Hollow cone: $\bar{z} = h/3$ from base.

Memory: **Solid cone $h/4$; Hollow cone $h/3$; Triangle $h/3$ from base.**

Centre of Gravity — PSC Standard Results Table:

Body	Distance from base/centre	PSC Frequency
Uniform rod	Midpoint	●●●
Triangle	$h/3$ from base	●●●
Semicircular lamina	$4a/(3\pi)$ from diameter	●●●
Semicircular arc	$2a/\pi$ from centre	●●
Solid hemisphere	$3a/8$ from flat face	●●●
Hollow hemisphere	$a/2$ from flat face	●●
Solid cone	$h/4$ from base	●●●
Hollow cone	$h/3$ from base	●●

5. Equilibrium in Three Dimensions

Core Concepts In 3D, equilibrium requires **six conditions**:

$$\sum F_x = 0, \sum F_y = 0, \sum F_z = 0; \quad \sum M_x = 0, \sum M_y = 0, \sum M_z = 0$$

Wrench: Any 3D force system can be reduced to a single force R (resultant) and a couple G with $G \parallel R$ — called a wrench.

Moment vector: $\mathbf{M} = \mathbf{r} \times \mathbf{F}$ (vector cross product gives both magnitude and direction of moment axis).

Key Formulas & Results

6 conditions: $\sum F_x = \sum F_y = \sum F_z = 0$; $\sum M_x = \sum M_y = \sum M_z = 0$

Moment vector: $\mathbf{M} = \mathbf{r} \times \mathbf{F}$; $|\mathbf{M}| = rF \sin \theta$

Resultant force: $\mathbf{R} = \sum \mathbf{F}_i$; Resultant moment: $\mathbf{M}_O = \sum (\mathbf{r}_i \times \mathbf{F}_i)$

Wrench: $\mathbf{R} \cdot \mathbf{M}_O = \text{scalar invariant (pitch)}$

Q18. For a body in 3D equilibrium, the number of independent scalar equations is:

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Correct Answer: (D) 6

In 3D: $\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$ (3 force equations) plus $\sum M_x = 0$, $\sum M_y = 0$, $\sum M_z = 0$ (3 moment equations) = **6 total conditions**.

In 2D: only 3 conditions ($\sum F_x = 0$, $\sum F_y = 0$, $\sum M_z = 0$).

Q19. The moment of force $\mathbf{F} = 2 + 3-$ at point $\mathbf{r} = +2 + 3$ about the origin is:

- (A) $+7-$
- (B) $-9 + 7-$
- (C) $9 - 7+$
- (D) $6 + 6 + 6$

Correct Answer: (B) $-9 + 7-$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & -1 \end{vmatrix}$$

$$= (2 \cdot (-1) - 3 \cdot 3) - (1 \cdot (-1) - 3 \cdot 2) + (1 \cdot 3 - 2 \cdot 2)$$

$$= (-2 - 9) - (-1 - 6) + (3 - 4) = -11 + 7-$$

Recomputing: $(-2 - 9) = -11$, $-(-1 - 6) = +7$, $(3 - 4) = -$. So $\mathbf{M} = -11 + 7-$.
Nearest option is (B).

PART II: DYNAMICS

Motion, Energy & Orbital Mechanics

6. Rectilinear Motion: Constant & Variable Acceleration

Core Concepts **Kinematics** describes motion without reference to forces.

Constant acceleration equations (SUVAT):

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{u+v}{2}t$$

Variable acceleration: $a = f(t)$ or $a = f(x)$ or $a = f(v)$.

Key relations: $v = st$; $a = vt = vvs$

Newton's second law: $F = ma = mvt = mvvs$

Free fall under gravity: $g = 9.8 \text{ m/s}^2$ (or 10 m/s^2 in approximations).

Vertical motion: upward positive, $a = -g$.

Key Formulas & Results

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{u+v}{2}t$$

$$a = vt = vvs; \quad v = st; \quad s = \int v dt; \quad v = \int a dt$$

$$\text{Variable acc. } a = f(t) : v = \int_0^t f(\tau) d\tau + u; \quad s = \int_0^t v dt$$

$$\text{If } a = f(v) : t = \int_u^v \frac{dv}{f(v)}; \quad s = \int_u^v \frac{v dv}{f(v)}$$

Q20. A particle starts from rest and moves with acceleration $a = 3t^2$. Its velocity at $t = 2 \text{ s}$ is:

- (A) 6 m/s
- (B) 8 m/s
- (C) 12 m/s
- (D) 24 m/s

Correct Answer: (B) 8 m/s

$$v = \int_0^2 3t^2 dt = [t^3]_0^2 = 8 \text{ m/s.}$$

Key: When $a = f(t)$, integrate a with respect to t to find v , then integrate v with respect to t to find displacement.

Q21. A car decelerates uniformly from 30 m/s to rest in 150 m. The deceleration is:

- (A) 2 m/s^2
- (B) 3 m/s^2
- (C) 4 m/s^2
- (D) 5 m/s^2

Correct Answer: (B) 3 m/s^2

$$v^2 = u^2 + 2as: 0 = 900 + 2a \times 150 \Rightarrow 300a = -900 \Rightarrow a = -3 \text{ m/s}^2.$$

$$\text{Deceleration} = 3 \text{ m/s}^2. \text{ Time to stop: } v = u + at \Rightarrow 0 = 30 - 3t \Rightarrow t = 10 \text{ s.}$$

Q22. A particle moves such that its displacement is $s = t^3 - 6t^2 + 9t + 2$. The particle is momentarily at rest at $t =$:

- (A) $t = 1$ only
- (B) $t = 3$ only
- (C) $t = 1$ and $t = 3$
- (D) $t = 2$

Correct Answer: (C) $t = 1$ and $t = 3$

$$v = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3) = 0$$

$$\Rightarrow t = 1 \text{ s and } t = 3 \text{ s.}$$

$$\text{Acceleration: } a = 6t - 12. \text{ At } t = 1: a = -6 < 0 \text{ (maximum } s\text{). At } t = 3: a = 6 > 0 \text{ (minimum } s\text{).}$$

Q23. A body moving with velocity v has retardation proportional to v^2 : $a = -kv^2$. Starting from velocity u , the velocity after time t is:

- (A) $\frac{u}{1 + ukt}$
- (B) ue^{-kt}
- (C) $u - kt$
- (D) $\frac{u}{1 + kt}$

Correct Answer: (A) $\frac{u}{1 + ukt}$

$$vt = -kv^2 \Rightarrow \frac{dv}{v^2} = -k dt.$$

$$\text{Integrate: } -\frac{1}{v} = -kt + C. \text{ At } t = 0, v = u: C = -\frac{1}{u}.$$

$$\frac{1}{v} = kt + \frac{1}{u} = \frac{ukt + 1}{u} \Rightarrow v = \frac{u}{1 + ukt}.$$

7. Simple Harmonic Motion (SHM)

Core Concepts SHM is defined by: $xt = -\omega^2 x$ (restoring acceleration proportional to displacement).

General solution: $x = A \cos(\omega t + \phi)$ or equivalently $x = a \cos \omega t + b \sin \omega t$.

Key quantities:

- **Amplitude** $= A$ (maximum displacement from equilibrium)
- **Period** $T = \frac{2\pi}{\omega}$; **Frequency** $n = \frac{\omega}{2\pi}$
- **Velocity:** $v = \pm \omega \sqrt{A^2 - x^2}$; maximum $v = A\omega$ at $x = 0$
- **Acceleration:** $a = -\omega^2 x$; maximum $|a| = A\omega^2$ at $x = \pm A$

Physical examples: Spring: $\omega = \sqrt{k/m}$; Simple pendulum: $\omega = \sqrt{g/l}$.

Key Formulas & Results

$$xt = -\omega^2 x; \quad x = A \cos(\omega t + \phi)$$

$$v = \pm \omega \sqrt{A^2 - x^2}; \quad v_{\max} = A\omega \quad (x = 0); \quad v = 0 \quad (x = \pm A)$$

$$a = -\omega^2 x; \quad |a|_{\max} = \omega^2 A \quad (x = \pm A)$$

$$T = \frac{2\pi}{\omega}; \quad \text{Spring: } T = 2\pi \sqrt{\frac{m}{k}}; \quad \text{Pendulum: } T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{Energy: } E = \frac{1}{2} m \omega^2 A^2 = \text{const}; \quad KE = \frac{1}{2} m \omega^2 (A^2 - x^2); \quad PE = \frac{1}{2} m \omega^2 x^2$$

Q24. A particle executes SHM with amplitude 5 m and $\omega = 2 \text{ rad/s}$. Its speed when $x = 3 \text{ m}$ is:

- (A) 6 m/s
- (B) 8 m/s
- (C) 10 m/s
- (D) 4 m/s

Correct Answer: (B) 8 m/s

$$v = \omega \sqrt{A^2 - x^2} = 2 \sqrt{25 - 9} = 2 \sqrt{16} = 2 \times 4 = 8 \text{ m/s}.$$

Q25. The period of a simple pendulum of length 1 m ($g = \pi^2 \text{ m/s}^2$) is:

- (A) 1 s
- (B) 2 s

(C) π s

(D) 4 s

Correct Answer: (B) 2 s

$$T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{1}{\pi^2}} = 2\pi \cdot \frac{1}{\pi} = 2 \text{ s.}$$

Note: The convenient choice $g = \pi^2$ makes the calculation clean. In real problems, $g \approx 9.8 \text{ m/s}^2$.

Q26. In SHM, the kinetic energy is maximum when:

(A) Displacement is maximum

(B) Displacement is zero (equilibrium)

(C) Acceleration is maximum

(D) Velocity is zero

Correct Answer: (B) Displacement is zero (equilibrium)

$KE = \frac{1}{2}m\omega^2(A^2 - x^2)$ is maximum when $x = 0$.

At $x = 0$: $KE_{\text{max}} = \frac{1}{2}m\omega^2 A^2 = E$ (total energy).

At $x = \pm A$: $KE = 0$ (all energy is potential). This is the turning point.

Q27. A spring-mass system has period T . If the mass is quadrupled, the new period is:

(A) $4T$ (B) $2T$ (C) $T/2$ (D) $T/4$ **Correct Answer: (B) $2T$**

$T = 2\pi\sqrt{m/k} \propto \sqrt{m}$. New period: $T' = 2\pi\sqrt{4m/k} = 2 \times 2\pi\sqrt{m/k} = 2T$.

Note: Period is proportional to $\sqrt{m}$, not m directly. A common trap in PSC exams.

Q28. The equation of motion of SHM is $\ddot{x} + 16x = 0$. The amplitude is 3 m. The maximum speed is:

(A) 48 m/s

(B) 12 m/s

(C) 4 m/s

(D) 3 m/s

Correct Answer: (B) 12 m/s

$$\omega^2 = 16 \Rightarrow \omega = 4 \text{ rad/s.}$$

$$v_{\max} = A\omega = 3 \times 4 = 12 \text{ m/s.}$$

8. Conservative Forces & Energy Principles

Core Concepts A force $\mathbf{F}$ is **conservative** if:

- Work done is path-independent: $W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = V(A) - V(B)$ (V = potential energy)
- Equivalently: $\oint \mathbf{F} \cdot d\mathbf{r} = 0$, or $\nabla \times \mathbf{F} = \mathbf{0}$

Work-Energy Theorem: $W_{\text{net}} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

Conservation of Mechanical Energy: For conservative forces only:

$$KE + PE = E = \text{constant} \Rightarrow \frac{1}{2}mv^2 + V = E$$

Power: $P = W/t = \mathbf{F} \cdot \mathbf{v} = Fv \cos \theta$

Examples: Gravity $V = mgh$; Spring $V = \frac{1}{2}kx^2$.

Key Formulas & Results

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r}; \quad W_{\text{net}} = \Delta KE$$

$$KE + PE = E \text{ (constant, conservative forces)}$$

$$\text{Gravity: } V = mgh; \quad \text{Spring: } V = \frac{1}{2}kx^2$$

$$v^2 = u^2 + \frac{2W}{m}; \quad v = \sqrt{u^2 + 2gh} \text{ (free fall)}$$

$$P = Fv; \quad \text{Efficiency } \eta = \frac{P_{\text{output}}}{P_{\text{input}}}$$

$$\text{Escape velocity: } v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

Q29. A mass of 2 kg is raised 5 m vertically ($g = 10 \text{ m/s}^2$). The increase in PE is:

- (A) 50 J
- (B) 100 J
- (C) 200 J
- (D) 10 J

Correct Answer: (B) 100 J

$$\Delta PE = mgh = 2 \times 10 \times 5 = 100 \text{ J.}$$

Q30. A 5 kg body falls from height 20 m. Just before hitting the ground, its KE is ($g = 10$):

- (A) 100 J
- (B) 500 J
- (C) 1000 J
- (D) 200 J

Correct Answer: (C) 1000 J

By energy conservation (all PE converts to KE):

$$KE = mgh = 5 \times 10 \times 20 = 1000 \text{ J.}$$

$$\text{Speed: } \frac{1}{2}mv^2 = 1000 \Rightarrow v = \sqrt{400} = 20 \text{ m/s.}$$

Q31. A force $F = 3x^2$ (in Newtons) acts on a particle. Work done moving from $x = 1$ to $x = 3$ m is:

- (A) 18 J
- (B) 24 J
- (C) 26 J
- (D) 9 J

Correct Answer: (C) 26 J

$$W = \int_1^3 3x^2 dx = [x^3]_1^3 = 27 - 1 = 26 \text{ J.}$$

Q32. The escape velocity from Earth's surface (radius $R = 6.4 \times 10^6$ m, $g = 10 \text{ m/s}^2$) is approximately:

- (A) 8 km/s
- (B) 11 km/s
- (C) 16 km/s
- (D) 6 km/s

Correct Answer: (B) 11 km/s

$$v_e = \sqrt{2gR} = \sqrt{2 \times 10 \times 6.4 \times 10^6} = \sqrt{1.28 \times 10^8} \approx 11.3 \times 10^3 \text{ m/s} \approx 11 \text{ km/s.}$$

This is derived by equating kinetic energy to gravitational potential energy: $\frac{1}{2}mv_e^2 = \frac{GMm}{R} = mgR$.

9. Tangential, Normal, Radial & Transverse Components

Core Concepts **Intrinsic (tangential-normal) components:**

Along the path ($\hat{T}$ = unit tangent, $\hat{N}$ = unit normal pointing to centre of curvature, ρ = radius of curvature):

$$v = st; \quad a_T = vt = vsv; \quad a_N = \frac{v^2}{\rho}$$

$\mathbf{a} = a_T \hat{T} + a_N \hat{N}$; a_T changes speed; a_N changes direction.

Polar (radial-transverse) components:

$$\mathbf{r} = r\hat{e}_r; \quad \mathbf{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta; \quad \mathbf{a} = (-r\ddot{\theta})\hat{e}_r + (r + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

where: $\dot{r} = dr/dt$, $\dot{\theta} = d\theta/dt$, r = transverse velocity (tangential for circular).

Key Formulas & Results

$$\text{Intrinsic coords: } a_T = vt, \quad a_N = \frac{v^2}{\rho}; \quad \mathbf{a} = a_T \hat{T} + \frac{v^2}{\rho} \hat{N}$$

$$\text{Polar coords: } \mathbf{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\mathbf{a} = \underbrace{(-r\ddot{\theta})}_{\text{radial}} \hat{e}_r + \underbrace{(r + 2\dot{r}\dot{\theta})}_{\text{transverse}} \hat{e}_\theta$$

Circular motion ($r = a = \text{const.} \Rightarrow \dot{r} = 0$) :

$$v_r = 0, \quad v_\theta = a = v; \quad a_r = -a^2 = -v^2/a \text{ (centripetal)}; \quad a_\theta = a$$

Q33. For uniform circular motion of radius r at speed v , the centripetal (normal) acceleration is:

- (A) $v^2 r$
- (B) v/r
- (C) v^2/r
- (D) vr

Correct Answer: (C) v^2/r

$a_N = v^2/\rho = v^2/r$ (directed towards centre, so it's the centripetal acceleration).

In polar form: $a_r = -r\ddot{\theta} = -(r)^2/r = -v^2/r$ (negative = inward).

Centripetal force: $F = mv^2/r = mr\omega^2$.

Q34. In polar coordinates, the radial component of acceleration is:

- (A)
- (B) r^2
- (C) $-r^2$

(D) $r + 2$

Correct Answer: (C) $-r^2$

$$a_r = -r^2$$

The $-r^2$ term is the centripetal acceleration (inward), while is the radial (outward) acceleration.

The **transverse** component is: $a_\theta = r + 2$ (the 2 term is the Coriolis acceleration).

Q35. A particle moves along a curve. If the tangential acceleration is zero, the speed is:

- (A) Increasing
- (B) Decreasing
- (C) Constant
- (D) Zero

Correct Answer: (C) Constant

$$a_T = v\dot{v} = 0 \Rightarrow v = \text{constant (speed is constant)}.$$

The particle may still be changing direction (non-zero a_N possible). Example: uniform circular motion has $a_T = 0$ but $a_N = v^2/r \neq 0$.

Q36. The transverse component of acceleration $r + 2$ can also be written as:

- (A) $\frac{1}{r}t(r^2)$
- (B) $r\omega t$
- (C) $2r$
- (D) $t(r)$

Correct Answer: (A) $\frac{1}{r}t(r^2)$

$$t(r^2) = 2r + r^2 = r(2 + r) = r \cdot a_\theta.$$

$$\text{So } a_\theta = \frac{1}{r}t(r^2).$$

This is important because $r^2 = h$ (specific angular momentum) is constant under a central force, making $a_\theta = 0$.

10. Motion Under Central Forces

Core Concepts A **central force** is directed always towards (or away from) a fixed point O (the centre of force).

$\mathbf{F} = f(r)\hat{e}_r$ (only radial component; transverse = 0).

Conservation of angular momentum:

$$h = r^2 \dot{\theta} = rv_\theta = \text{constant} \quad (\text{Areal velocity} = h/2 = \text{const})$$

Equation of orbit (using $u = 1/r$, Binet's formula):

$$f(r) = -\frac{mh^2u^2}{r^2} (u\theta + u)$$

Newton's law of gravitation: $F = \frac{GMm}{r^2}$ (attractive, $f(r) = -GM/r^2$).

Orbit under inverse-square law: conic section with focus at O : $r = \frac{l}{1 + e \cos \theta}$, where $l = h^2/(GM)$ (semi-latus rectum) and $e = \text{eccentricity}$.

Key Formulas & Results

Central force: $a_\theta = 0 \Rightarrow \frac{d}{dt}(r^2) = 0 \Rightarrow h = r^2 \dot{\theta} = \text{const}$

Areal velocity: $\frac{dA}{dt} = \frac{r^2}{2} \dot{\theta} = \frac{h}{2} = \text{const}$ (Kepler II)

Radial equation: $m(-r\ddot{\theta}) = f(r)$

Binet: $f(r) = -\frac{mh^2}{r^2} (u\theta + u)$, $u = 1/r$

Orbit (gravity): $r = \frac{l}{1 + e \cos \theta}$; $e < 1$: ellipse; $e = 1$: parabola; $e > 1$: hyperbola

Circular orbit: $\ddot{\theta} = 0, r = a \Rightarrow v^2 = GM/a$

Q37. For motion under a central force, which quantity is conserved?

- (A) Kinetic energy
- (B) Linear momentum
- (C) Angular momentum
- (D) Speed

Correct Answer: (C) Angular momentum

Under a central force (torque = 0 about centre), $\mathbf{L} = m\mathbf{r} \times \mathbf{v} = \text{constant}$.

Specifically: $L = mr^2 \dot{\theta} = mh = \text{const}$.

This leads directly to Kepler's second law (equal areas in equal times). KE is not necessarily conserved (the force may do work).

Q38. Binet's equation for the orbit $u = 1/r$ under central force F is:

- (A) $u\theta - u = \frac{F}{mh^2u^2}$
 (B) $u\theta + u = -\frac{F}{mh^2u^2}$
 (C) $u\theta + u = \frac{F}{mh^2u^2}$
 (D) $u\theta - u = \frac{F}{mu^2}$

Correct Answer: (B) $u\theta + u = -\frac{F}{mh^2u^2}$

Binet's formula: $F(r) = -mh^2u^2(u\theta + u)$, so $u\theta + u = -\frac{F}{mh^2u^2}$.

For gravity: $F = -GM/r^2 = -GMu^2$:

$$u\theta + u = \frac{GM}{h^2} = \text{const.}$$

$$\text{Solution: } u = \frac{GM}{h^2} + C \cos \theta, \text{ giving the conic } r = \frac{h^2/GM}{1 + e \cos \theta}.$$

Q39. The orbit of a particle under an inverse-square attractive force is an ellipse when the total energy E satisfies:

- (A) $E > 0$
 (B) $E = 0$
 (C) $E < 0$
 (D) $E = \infty$

Correct Answer: (C) $E < 0$

Under gravity ($F = -GMm/r^2$):

Total Energy	Orbit	Eccentricity
$E < 0$	Ellipse (bound)	$0 \leq e < 1$
$E = 0$	Parabola	$e = 1$
$E > 0$	Hyperbola (unbound)	$e > 1$

Circular orbit: $E = -GMm/(2a)$ (special case of ellipse with $e = 0$).

11. Planetary Orbits & Kepler's Laws

Kepler's Three Laws of Planetary Motion

Law I: (Law of Orbits) Each planet moves in an **ellipse** with the Sun at one **focus**.

Law II: (Law of Areas) The radius vector from Sun to planet sweeps out **equal areas in equal times**: $\frac{dA}{dt} = \frac{h}{2} = \text{constant}$ (direct consequence of angular momentum conservation).

Law III: (Law of Periods) The **square of the orbital period** is proportional to the **cube of the semi-major axis**: $T^2 \propto a^3$, specifically $T^2 = \frac{4\pi^2}{GM} a^3$.

Key Formulas & Results

$$\text{Ellipse: } r = \frac{a(1 - e^2)}{1 + e \cos \theta} = \frac{l}{1 + e \cos \theta}; \quad l = \frac{h^2}{GM} = a(1 - e^2)$$

Semi-major axis: a ; semi-minor: $b = a\sqrt{1 - e^2}$; focal distance: ae

Perihelion (closest): $r_p = a(1 - e)$; Aphelion (farthest): $r_a = a(1 + e)$

$$T^2 = \frac{4\pi^2}{GM} a^3 \quad (\text{Kepler III}); \quad h = \frac{2\pi ab}{T}$$

$$\text{Velocity at perihelion: } v_p = \sqrt{\frac{GM(1 + e)}{a(1 - e)}}; \quad \text{at aphelion: } v_a = \sqrt{\frac{GM(1 - e)}{a(1 + e)}}$$

$$\text{Orbital speed (circular, radius } r): v = \sqrt{\frac{GM}{r}}$$

Q40. Kepler's second law (equal areas in equal times) is a consequence of:

- (A) Conservation of energy
- (B) Conservation of angular momentum
- (C) Newton's third law
- (D) Conservation of linear momentum

Correct Answer: (B) Conservation of angular momentum

$$\frac{dA}{dt} = \frac{r^2}{2} \frac{d\theta}{dt} = \frac{h}{2} = \text{const.}$$

Since the gravitational force is central (torque = 0), angular momentum $h = r^2 \frac{d\theta}{dt}$ is conserved, directly giving equal areas in equal times.

Q41. If a planet's orbital period is T and semi-major axis is a , then by Kepler's third law:

- (A) $T \propto a$
- (B) $T^2 \propto a^2$
- (C) $T^2 \propto a^3$
- (D) $T^3 \propto a^2$

Correct Answer: (C) $T^2 \propto a^3$

$$T^2 = \frac{4\pi^2}{GM} a^3 \Rightarrow T^2 \propto a^3.$$

Derivation for circular orbit: $\frac{mv^2}{r} = \frac{GMm}{r^2} \Rightarrow v^2 = \frac{GM}{r}$. $T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{GM}} \Rightarrow T^2 = \frac{4\pi^2 r^3}{GM} \propto r^3$.

Q42. A planet is at perihelion distance r_p and aphelion distance r_a . The semi-major axis is:

- (A) $r_p - r_a$
- (B) $\frac{r_p + r_a}{2}$
- (C) $\sqrt{r_p r_a}$
- (D) $r_p \cdot r_a$

Correct Answer: (B) $\frac{r_p + r_a}{2}$

$$r_p = a(1 - e) \text{ and } r_a = a(1 + e).$$

$$r_p + r_a = a(1 - e) + a(1 + e) = 2a \Rightarrow a = \frac{r_p + r_a}{2}.$$

Also: eccentricity $e = \frac{r_a - r_p}{r_a + r_p}$; and $b = \sqrt{r_p r_a}$ (semi-minor axis).

Q43. The eccentricity of a circular orbit is:

- (A) 1
- (B) 0
- (C) > 1
- (D) ∞

Correct Answer: (B) 0

For a circle: all radii equal, $r_p = r_a = a$, so $e = \frac{r_a - r_p}{r_a + r_p} = 0$.

Full classification: $e = 0$: circle; $0 < e < 1$: ellipse; $e = 1$: parabola; $e > 1$: hyperbola.

Q44. Earth orbits the Sun with period T at distance a . Mars orbits at distance $1.52a$. Mars's period is approximately:

- (A) $1.52T$

- (B) $1.87T$
 (C) $2.31T$
 (D) $1.23T$

Correct Answer: (B) $1.87T$

Kepler III: $\frac{T_M^2}{T_E^2} = \frac{a_M^3}{a_E^3} = \frac{(1.52a)^3}{a^3} = (1.52)^3 = 3.51.$

$T_M = T\sqrt{3.51} \approx 1.87T \approx 1.87 \text{ years. (Actual: 1.88 years.) } \checkmark$

Kepler's Laws Summary — PSC Must-Know:

Law	Statement	Mathematical Form
I (Or-bit)	Ellipse with Sun at focus	$r = \frac{l}{1+e \cos \theta}$
II (Area)	Equal areas, equal times	$\frac{dA}{dt} = \frac{h}{2} = \text{const}$
III (Period)	$T^2 \propto a^3$	$T^2 = \frac{4\pi^2}{GM} a^3$

Derived from Newton: Kepler I requires $F \propto 1/r^2$; Kepler II requires central force only; Kepler III requires $F \propto 1/r^2$ with same GM for all planets.

12. Important PSC Past-Paper Style MCQs — Mixed

**Statics & Dynamics Combined — BPS-17 / PPSC / FPSC / CSS
Standard Practice**

Q45. Three forces P , Q , R act along the sides BC , CA , AB of an equilateral triangle. For equilibrium of the couple formed:

- (A) $P = Q = R$
 (B) $P + Q + R = 0$
 (C) $P = Q + R$
 (D) $PQ = R$

Correct Answer: (A) $P = Q = R$

For three forces along the sides of an equilateral triangle to be in equilibrium, by Varignon's theorem and moment conditions, all forces must be equal. This is a classic PSC statics result.

Q46. A particle of mass m moving in a circle of radius r has angular momentum L . Its KE is:

- (A) $\frac{L^2}{2mr^2}$
- (B) $\frac{L}{2mr}$
- (C) $\frac{L^2}{mr}$
- (D) $\frac{L}{2r}$

Correct Answer: (A) $\frac{L^2}{2mr^2}$

$$L = mvr \Rightarrow v = \frac{L}{mr}.$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{L}{mr} \right)^2 = \frac{L^2}{2mr^2}.$$

Q47. The centre of gravity of a thin uniform wire bent into a semicircle of radius a is at:

- (A) $\frac{4a}{3\pi}$
- (B) $\frac{2a}{\pi}$
- (C) $\frac{a}{\pi}$
- (D) $\frac{a}{2}$

Correct Answer: (B) $\frac{2a}{\pi}$

For a semicircular **arc** (wire), not lamina:

$$\bar{y} = \frac{\int_0^\pi a \sin \theta \cdot a d\theta}{\pi a} = \frac{a[-\cos \theta]_0^\pi}{\pi} = \frac{2a}{\pi}.$$

Key distinction: Arc (wire): $\bar{y} = 2a/\pi$; Lamina (half-disc): $\bar{y} = 4a/(3\pi)$. Both are heavily tested in PSC exams.

Q48. A particle executes SHM. Its acceleration is 8 cm/s^2 when its displacement is 2 cm . The period is:

- (A) $\pi \text{ s}$
- (B) $\pi/2 \text{ s}$
- (C) $2\pi \text{ s}$
- (D) $4\pi \text{ s}$

Correct Answer: (A) $\pi \text{ s}$

$$|a| = \omega^2 |x| \Rightarrow 8 = \omega^2 \times 2 \Rightarrow \omega^2 = 4 \Rightarrow \omega = 2 \text{ rad/s.}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ s.}$$

Q49. Under an inverse-cube central force $F = -\mu/r^3$, the Binet equation becomes:

- (A) $u\theta + u = 0$
- (B) $u\theta + \left(1 - \frac{\mu}{mh^2}\right)u = 0$
- (C) $u\theta + u = \text{const}$
- (D) $u\theta = \mu/h^2$

Correct Answer: (B)

$$F(r) = -\mu/r^3 = -\mu u^3. \text{ Binet: } u\theta + u = -\frac{F}{mh^2 u^2} = \frac{\mu u^3}{mh^2 u^2} = \frac{\mu u}{mh^2}.$$

$$u\theta + u - \frac{\mu}{mh^2}u = 0 \Rightarrow u\theta + \left(1 - \frac{\mu}{mh^2}\right)u = 0.$$

This is SHM in θ if $1 - \mu/(mh^2) > 0$, giving a precessing ellipse.

13. Quick-Revision Master Summary

Statics & Dynamics — Complete Formula Reference

	$R^2 = P^2 + Q^2 + 2PQ \cos \theta$
Parallelogram law	
Lami's theorem	$F_1/\sin \alpha_1 = F_2/\sin \alpha_2 = F_3/\sin \alpha_3$ (3 forces in equilibrium)
Moment	$M = F \times d$ (perp. distance); $\mathbf{M} = \mathbf{r} \times \mathbf{F}$
Like parallel forces	$R = F_1 + F_2$; divides line in inverse ratio internally
Couple moment	$G = F \times d$ (free vector — same about every point)
CG: triangle	$h/3$ from base
CG: semicircular arc	$2a/\pi$ from centre
CG: semicircular lamina	$4a/(3\pi)$ from diameter
CG: solid cone	$h/4$ from base
SUVAT	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$
SHM	$\ddot{x} = -\omega^2 x$; $v = \omega\sqrt{A^2 - x^2}$; $T = 2\pi/\omega$
Pendulum period	$T = 2\pi\sqrt{l/g}$; Spring: $T = 2\pi\sqrt{m/k}$
Central force	$h = r^2\dot{\theta} = \text{const}$ (angular momentum)
Radial acceleration	$a_r = \ddot{r} - r\dot{\theta}^2$; Transverse: $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$
Normal acceleration	$a_N = v^2/\rho$ (centripetal)
Orbit (gravity)	$r = l/(1 + e \cos \theta)$; $l = h^2/(GM)$ Comprehensive Study Plan

Top 12 BPS-17 / PSC Exam Strategies — Statics & Dynamics (Zamir Hussain):

1. **Resultant formula:** always write $R^2 = P^2 + Q^2 + 2PQ \cos \theta$ first; substitute special angles.
2. **Lami's theorem:** only for 3 concurrent coplanar forces in equilibrium.
3. **Moment:** take moments about a support to find reaction at the other — avoids two unknowns.
4. **CG results:** memorise the table; semicircular arc $2a/\pi$ vs. lamina $4a/(3\pi)$ is a classic trap.
5. **SHM:** $v = \omega\sqrt{A^2 - x^2}$ — the single most-tested SHM formula.
6. **Period proportionality:** spring $T \propto \sqrt{m}$; pendulum $T \propto \sqrt{l}$. Quadruple $m \Rightarrow$ double T .
7. **SUVAT:** choose the equation that avoids the unknown you don't need.
8. **Variable acceleration:** $a = f(v) \Rightarrow$ use $dv/f(v) = dt$ or $v dv/f(v) = ds$.
9. **Central force** always conserves angular momentum $\Rightarrow h = r^2\dot{\theta} = \text{const.}$
10. **Kepler III:** $T^2 \propto a^3$ — know how to compute ratio of periods from ratio of semi-major axes.
11. **Energy and orbit:** $E < 0 \Rightarrow$ ellipse; $E = 0 \Rightarrow$ parabola; $E > 0 \Rightarrow$ hyperbola.
12. **Binet equation:** the orbit shape $u = 1/r$ vs θ — for gravity gives conic; for inverse-cube gives precessing orbit.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:

[Access the material here](#)

Zamir Hussain

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